Identification of Three-Nucleon Short-Range Correlations in Electron Scattering

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Three-nucleon short-range correlations (3N-SRCs) are a fundamental, yet experimentally unestablished, component of nuclear structure. We develop a framework to isolate 3N-SRCs in electron scattering by extending the Generalized Contact Formalism to three-nucleon systems and implementing it in an event generator. The calculations reveal distinct momentum-space topologies, including asymmetric "rocket" and symmetric "star" configurations with different kinematic signatures. Contributions at $x_B > 2$ are dominated by asymmetric configurations, while symmetric configurations are accessed near $x_B \approx 1$. Backgrounds from two-nucleon SRCs with final-state interactions are quantified using transport simulations and shown to exhibit separable angular correlations. We identify kinematic selections and observables that enhance sensitivity to 3N-SRCs and enable data-driven signal extraction, providing guidance for future high-precision exclusive and semi-exclusive measurements.

I. INTRODUCTION

Short-range correlations (SRCs) among nucleons provide a direct window into the high-momentum components of nuclear structure and the dynamics of strongly interacting many-body systems [1–3]. Two-nucleon short-range correlations (2N-SRCs) have been extensively investigated in inclusive, semi-inclusive, and exclusive electron-scattering experiments [4–11], establishing that nucleon pairs with large relative momenta and small spatial separations constitute an important component of nuclear wave functions. In contrast, three-nucleon short-range correlations (3N-SRCs) remain considerably less well understood, both experimentally and theoretically.

3N-SRCs are expected to arise when three nucleons simultaneously probe sufficiently short distances that their relative momenta extend into the high-momentum region of the nuclear wave function. At such distances, the short-range part of the nucleon-nucleon interaction, as well as possible contributions from three-nucleon forces, becomes increasingly important. Three-body short-range configurations therefore provide a unique probe of nuclear dynamics at short distances and high density, beyond the regime in which the nuclear system can be adequately described in terms of predominantly independent nucleons or correlated pairs. Establishing the existence and properties of 3N-SRCs is thus important for a complete understanding of nuclear structure and may provide insight into the behavior of strongly interacting matter at high density, with implications extending to the physics of neutron stars.

Despite their importance, an unambiguous experimental identification of 3N-SRCs remains elusive. Inclusive electron scattering at Bjorken $x_B > 2$ has been proposed as a promising approach because this region kinematically suppresses scattering from stationary single nucleons and enhances sensitivity to multi-nucleon configurations [4, 12, 13]. However, the interpretation of the measured strength is complicated by several competing mechanisms. In particular, final-state interactions (FSI), the center-of-mass (c.m) motion of correlated nucleon pairs, and contributions from 2N-SRC configurations can all generate strength in the same kinematic region and can therefore obscure or mimic a genuine 3N-SRC signal. Consequently, an enhancement of the cross section at $x_B > 2$, by itself, does not provide an unambiguous identification of 3N-SRCs.

Semi-inclusive and exclusive measurements offer additional handles on the underlying configurations through the detection of one or more correlated nucleons. Triple-coincidence measurements, in particular, can provide substantially more direct information on three-nucleon configurations. However, these measurements introduce additional experimental and theoretical challenges associated with detector acceptance, multidimensional kinematics, FSI, reaction mechanisms, and the description of the correlated nuclear wave function. The central experimental challenge is therefore not simply to observe strength at large x_B or high momentum, but to demonstrate that the observed strength contains a component that cannot be accounted for by 2N-SRCs, FSI, and the c.m motion of correlated pairs.

Momentum transfer provides an important additional handle on this problem. As Q^2 increases, the spatial resolution of the electron probe improves, allowing progressively shorter-distance configurations of the nuclear system to be resolved. This increasing resolution is expected to enhance sensitivity to the high-momentum components associated with SRCs and, potentially, to configurations involving three nucleons at short distances. The relevant experimental question is therefore not only whether 3N-SRCs contribute to electron-scattering observables, but at what momentum transfer and nuclear momentum their contribution becomes distinguishable from the substantially larger backgrounds associated with 2N-SRCs and FSI.

Recent theoretical and experimental developments make it possible to address this question quantita-

tively. Ab-initio many-body calculations and contact-based approaches have provided new insight into the high-momentum components of nuclear wave functions and the structure of short-distance correlations. In particular, the generalized contact formalism (GCF) provides a framework for relating high-momentum nuclear configurations to short-distance two- and three-body contacts and for calculating their contributions to experimentally accessible observables. At the same time, the high luminosity, large acceptance, and advanced detection capabilities of modern electron-scattering facilities, particularly Jefferson Lab, make measurements at high momentum transfer and large nuclear momentum increasingly feasible.

In this work, we use the generalized contact formalism extended to three-nucleon short-range correlations to quantitatively investigate the prospects for identifying 3N-SRCs in electron scattering at kinematics accessible at Jefferson Lab. We evaluate the expected 3N-SRC contribution for 12C nuclei and examine its separation from backgrounds associated with 2N-SRCs, final-state interactions, and the center-of-mass motion of correlated pairs. We investigate how the sensitivity to genuine three-nucleon configurations evolves with momentum transfer and nuclear momentum and identify kinematic regions in which the 3N-SRC contribution is expected to become experimentally distinguishable from these competing mechanisms. Our results provide initial quantitative guidance for the design and interpretation of future measurements aimed at establishing the existence, strength, and structure of three-nucleon short-range correlations in nuclei.

II. THE KINEMATICAL CONFIGURATION OF 3N-SRC CLUSTERS IN MOMENTUM SPACE

Short-range correlations (SRCs) involving two nucleons (2N-SRC) are traditionally defined as pairs of nucleons with momenta significantly above the Fermi momentum and with a correspondingly low center-of-mass (c.m.) momentum. In this regime, the geometry of the configuration is relatively well understood: the two high-momentum nucleons predominantly appear in a back-to-back arrangement, reflecting their large relative momentum and small c.m. motion.

By analogy, we define three-nucleon short-range correlations (3N-SRC) as configurations in which all three nucleons carry momenta above the Fermi sea while the total c.m. momentum of the triplet remains small. Unlike the 2N-SRC case, the spatial and momentum geometry of 3N-SRC configurations is not uniquely determined. Several distinct arrangements of the three high-momentum nucleons can, in principle, satisfy the defining criteria of high individual momenta and low c.m. momentum.

In the discussion below, we outline the possible momentum-space configurations that may contribute to 3N-SRCs and examine how they differ dynamically and geometrically from the well-characterized 2N-SRC case.

III. SIMULATING 3N-SRC

The **Generalized Contact Formalism (GCF)** provides a factorized theoretical framework for describing high-momentum nucleons in nuclei arising from **two-nucleon short-range correlations (2N-SRC)**. The GCF was derived and validated against many-body Quantum Monte Carlo calculations in Refs. [14–16] and was successfully applied to the analysis of electron scattering SRC measurements in Refs. [17–20]. Here we extend the GCF to describe the 3N-SRC clusters and we will use the GCF predictions as a guideline to the search for 3N-SRC signature in electron scattering data.

In the GCF the factorization separates the **short-distance, high-momentum nucleon dynamics**, which are largely nucleus-independent, from the **long-range structural information** encoded in the contacts. The electron-nucleus cross section using a factorization of the electron-nucleon part from the nuclear spectral function is given by

$$\frac{d^6\sigma}{d^2\Omega_k\, dE_k\, d^3\vec{\boldsymbol{p}}'_{\mathbf{1}}} \;=\; \sigma_{e,N}\, S, \tag{1}$$

where $\sigma_{e,N}$ is the electron-nucleon cross section, E_k and Ω_k are the energy and solid angle of the scattered electron, $\vec{p}'_1$ the momentum of the struck nucleon, and S is the 3N spectral function.

We keep the basic assumption of contact theory (factorization) and express the 3N spectral function as a convolution of a 3-body local density $\rho^\alpha(\vec{p}_1,\vec{p}_2,\vec{p}_3)$ and the SRC density relative to the nucleus ($n^\alpha(\vec{p}_1+\vec{p}_2+\vec{p}_3)$). The spectral function must also be summed over the quantum numbers of the SRC such as spin and isospin which are both grouped into α. The contacts (C^α) effectively weight the probability of each possible SRC configuration.

Finally, we integrate over the second and third nucleon momenta to obtain the spectral function to be used in Eq. (1).

$$S \;=\; \frac{1}{4\pi}\sum_\alpha C^\alpha\, t^\alpha \int \frac{d^3\vec{p}_2\, d^3\vec{p}_3}{(2\pi)^6}\, \delta(f(\vec{p}_2,\vec{p}_3))\, \rho^\alpha(\vec{p}_1,\vec{p}_2,\vec{p}_3)\, n^\alpha(|\vec{p}_1+\vec{p}_2+\vec{p}_3|)\,. \tag{2}$$

t^α is a combinatorial factor for the number of protons/neutrons in the SRC. The 3N SRC density is a func-

tion of the 3 nucleon momenta, and the center of mass distribution is just sampling from a Gaussian distribution.

The 2 nucleon density functions come from solving the Schrödinger equation for a given potential between 2 nucleons and squaring the wavefunction. For the 3N case we used the Helium-3 ground state wavefunction as a substitute for the 3N SRC wavefunction:

$$\rho(p_a, p_b, \theta_{a,b}) \;=\; \left|\tilde{\varphi}^{\alpha}(p_a, p_b, \theta_{a,b})\right|^2 , \tag{3}$$

where p_a is the momentum of particle 1 in the center of mass frame, p_b is the relative momentum between particle 2 and 3, and the angle $\theta_{a,b}$ is the angle between the 2 vectors.

Changing variables using appropriate Jacobians the final expression for the cross section is:

$$\frac{d^{11}\sigma^{\alpha}}{d^2\Omega_k\, d^3\vec{p}_{cm}\, d^2\Omega_a\, d\phi_{b,a\|z}\, dp_a\, dp_b\, d\theta_{a,b}} \;=\; \frac{1}{2(2\pi)^7}\,\sigma_{e,N}\, C^{\alpha}\, t^{\alpha}\, \rho^{\alpha}(p_a, p_b, \theta_{a,b})\, n^{\alpha}(p_{cm}) \left|\frac{E_1'}{E_1' - \vec{p}_1' \cdot \hat{k}}\right| . \tag{4}$$

See supplementary material for more details.

We built an event generator with event weight based on the cross section in Eq. (4). We sample randomly from only 2 electron momentum variables, the last variable (E_k) was fully defined by conservation of energy. Events that did not fulfill that request were rejected. See supplementary material for more details. As presented below we also used the GCF to simulate 2N-SRC. Both event generators (2N-SRC and 3N-SRC) are then used in conjunction with GENIE [21, 22] to account for final-state interactions (FSI).

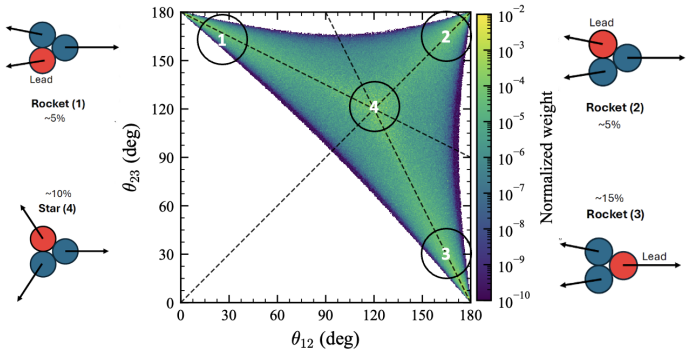


FIG. 1. 3N-SRC configurations. Shown are 3N-SRC events generated by the event generator discussed above (*without* accounting for FSI) and the selected geometries (configurations) with 3 high momentum nucleons in the final state. The nucleon marked online in red and called "leading" is the nucleon that absorbed the virtual photon. Not shown is a configuration similar to "Rocket (3)" with the two recoil nucleons replaced by a deuteron. The dashed lines indicate the symmetry axes, as determined by permutations of the three nucleons.

IV. RESULTS

A. 3N-SRC configurations

The 3N-SRC events produced by the event generator (without accounting for FSI) are shown in Fig. 1 as a function of the two opening angles between the high momentum 3 nucleons in the initial state. In red and marked as "leading" is the nucleon that absorbed the virtual photon. Any of the shown nucleons can be either a proton or a neutron.

In this work we will refer to four basic final-state configurations with three high momentum nucleons, and one with a high momentum pair and one high momentum nucleon.

The four basic configurations considered in this work are shown in Fig. 1 as a function of the opening angles between high-energy nucleons in the initial nuclear ground state. The configuration in which two nucleons are balanced by a third nucleon with higher momentum is referred to as the "rocket" configuration. In contrast, the configuration in which three nucleons emerge with similar high momenta and are symmetrically arranged with approximately 120° opening angles between each pair is referred to as the "star" configuration. The dashed lines intersecting at the center of the star configuration indicate the symmetry axes, as determined by permutations of the three nucleons.

The ground state momentum of the 3 nucleons in the configurations presented above is shown in Fig. 2. As can be seen in the figure the rocket configuration have a typical single nucleon with about 700-800 MeV/c balanced by two other nucleons with 400-600 MeV/c. The star configuration involves 3 nucleons with 500-700 MeV/c nucleons. We name the different rocket configurations by the type (lead or recoil) of the highest momentum nucleon.

The electron scattering reaction kinematics selectively populate different 3N-SRC configurations. For example $x_B > 2$ is possible only for the leading rocket configuration as can be seen in Figs. 3, 4. The star configuration on the other hand populates mainly $x_B \approx 1 \pm 0.5$.

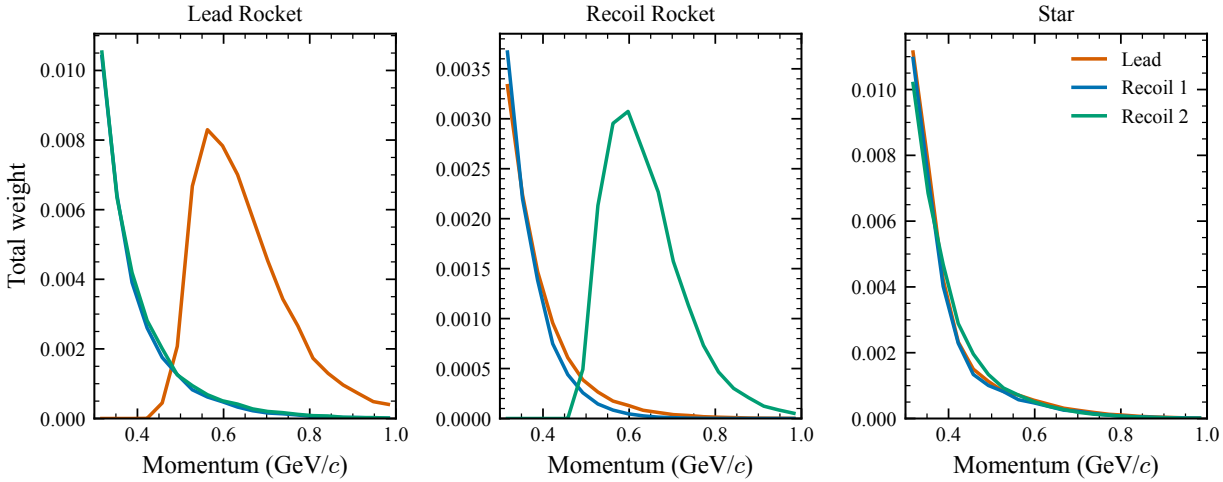


FIG. 2. The ground state momentum distribution of the 3 SRC nucleons in the rocket and star like configurations.

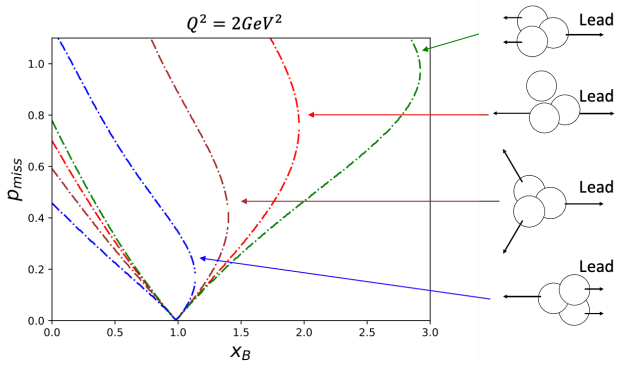


FIG. 3. The x_B vs. P_{miss} distributions for 3 typical 3N-SRC and 2N-SRC configurations.

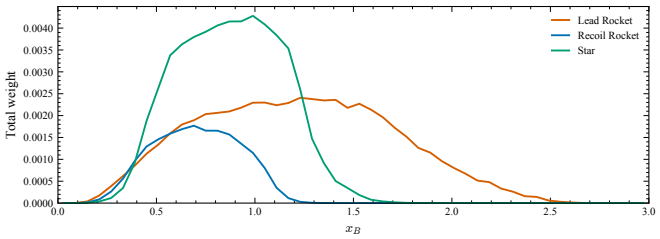


FIG. 4. The x_B distributions for 3 typical 3N-SRC configurations. As can be seen clearly the $x_B > 2$ 3N-SRC is populated with leading rocket configuration only.

B. 2N-SRC with FSI

In electron-induced $e, e'pp$ and $e, e'ppN$ measurements, the identification of 3N-SRC is based on the detection and reconstruction of two or three nucleons carrying high momenta attributed to the initial nuclear ground state. This interpretation is sensitive to final-state interactions (FSI), in which the outgoing nucleons rescatter from one another or from the residual nucleus. Such rescattering can modify the momenta and angular correlations of the detected nucleons, thereby distorting the reconstructed initial-state distributions. In particular, FSI can cause events originating from 2N-SRC to populate the kinematic regions associated with 3N-SRC, potentially leading to their misidentification as 3N-SRC events.

To minimize such interpretation complications, we use kinematical conditions and cuts that prefer the 3N-SRC state over contamination from non-3N-SRC initial states.

It is thus needed to study the impact of FSI on SRC measurements using theoretical approaches that are suitable for medium and heavy nuclei studies. One such approach uses effective transport theory, where effective nucleon-nucleon scattering cross-sections are used to calculate the propagation of nucleons as they traverse the nuclear medium.

Our calculations use the GCF to model the ground-state distribution of SRC pairs and triplets in nuclei [14–16], in conjunction with *GENIE*'s scattering event generator [21, 22]. The latter simulates the scattering process and accounts for FSI via a data-driven Intra Nuclear Cascade transport model [21, 23, 24].

Final-state interactions were simulated using the semi-classical intranuclear cascade model "GENIE hN" [24–26]. GENIE tracks each hadron produced in the initial lepton–nucleus interaction, as well as those generated during the cascade. At each step, the probability of reinteraction is determined by the relevant cross section and the local nuclear density along the hadron's path. If a reinteraction occurs, the model probabilistically selects the reaction type (e.g., charge exchange, inelastic scattering) according to the relative cross sections, and determines the resulting final-state particles and their kinematics. Each outgoing particle is then propagated independently.

We study 2N-SRC plus FSI and 3N-SRC plus FSI in the channel of events where two high-momentum protons are detected and a recoil high-momentum nucleon

is reconstructed. We assess the effectiveness of the kinematical cuts applied to identify true 3N-SRC events over 2N-SRC events and estimate the fraction of 2N-SRC that contaminates the 3N-SRC samples accounting for FSI. We do so for the nucleus ^{12}C. The 2N-SRC is the main source of background. We also discuss below the background contribution from mean field nucleons with FSI. See Fig. 7 and discussion.

In Fig. 5 (left), we present 3N-SRC plus FSI events obtained from the $^{12}\mathrm{C}(e, e'pp)$ simulation, with the third (neutron) momentum reconstructed from the missing momentum. All nucleons have momenta above the Fermi level. We apply typical selection cuts as used to identify the leading proton in SRC measurements [7–10]: the angle between the leading proton and $\vec{q}$ is less than 8°, and the leading proton momentum exceeds $0.75\,|\vec{q}|$. In addition, the missing momentum in the $(e, e'p_{\mathrm{lead}})$ channel is restricted to the range 0.25–0.9 GeV/c, and we require $x_B < 1.2$.

The distribution of 2N-SRC events with FSI, passing the same selection criteria as the 3N-SRC events, is also shown in Fig. 5 (right). As is evident, the event distributions for genuine 3N-SRC and background differ significantly, enabling the extraction of the signal-to-background (S/B) ratio from a measurement.

For example, the number of 3N-SRC+FSI events in the left and right triangular regions of Fig. 5 is skewed to the left (ratio ≈ 9:1). In contrast, the corresponding ratio for the 2N-SRC+FSI background differs substantially, being highly skewed to the right (ratio ≈ 1 : 100). Assuming, for instance, an overall 3N-SRC to 2N-SRC ratio of 10% within the left triangular region, the ratio of total events in the left corner to those in the right becomes approximately 0.1, as illustrated in Fig. 6. More generally, Fig. 6 can be used to extract the signal-to-background ratio and to determine the number of 3N-SRC events after background subtraction for any mixture.

To ensure a comprehensive evaluation of potential background sources, we also simulated single-nucleon knockout events using a local Fermi gas model coupled with the GENIE FSI transport model, applying the same requirement of detecting two high-momentum final-state nucleons with a recoild high-momentum nucleon being reconstructed. Observing the opening-angle distribution 7, the single-nucleon FSI background exhibits a strong spatial asymmetry, but with an even more pronounced skewness toward the right triangular region (with the left and right regions containing 0.06% and 24% of the events, respectively - four times larger than that observed for the 2N-SRC background), thus not interfering with the suggested method of S/B ratio extraction. Moreover, the overall contamination from this single-nucleon mechanism remains minimal; the total fraction of events surviving the applied SRC kinematic selections is smaller by a factor of 20 compared to the 2N-SRC background fraction.

SUMMARY

This paper addresses the challenge of identifying **three-nucleon short-range correlations (3N-SRCs)**, which are believed to play an important role in nuclear structure but have not yet been clearly established experimentally. We extend the **Generalized Contact Formalism (GCF)**-a framework previously used for two-nucleon systems-to describe interactions among three nucleons. We also implement this extended model in an event generator to simulate electron-nucleus scattering experiments.

Using these simulations, the paper predicts that 3N-SRCs produce distinct momentum configurations: asymmetric "rocket"-like structures and symmetric "star"-like patterns. These configurations appear in different kinematic regions, with asymmetric ones dominating at high Bjorken $x_B > 2$, while symmetric configurations are more accessible near $x_B \approx 1$.

The study also carefully examines competing background processes, particularly two-nucleon correlations combined with final-state interactions, and shows that these backgrounds have distinguishable angular patterns. Based on this, the authors propose specific kinematic cuts and observables that enhance the visibility of genuine 3N-SRC signals.

Overall, the paper provides ideas and practical guidance for future high-precision scattering experiments aimed at isolating and confirming the existence of three-nucleon correlations in nuclei using high energy electron beams as in Jefferson Lab.

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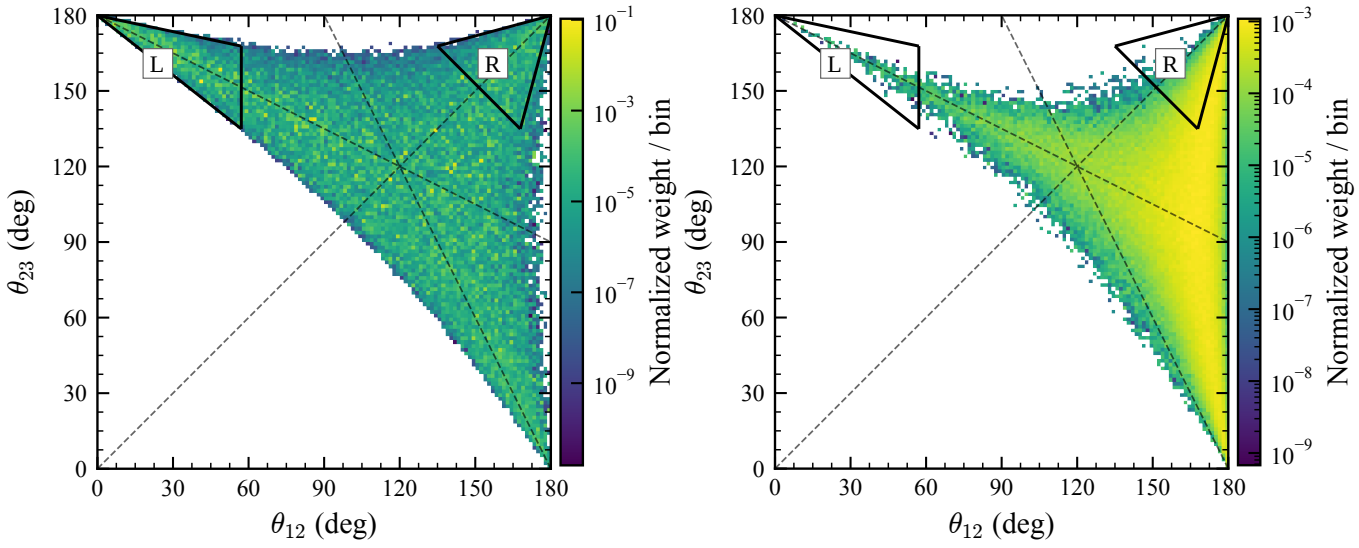


FIG. 5. 3N-SRC plus FSI events (left) and 2N-SRC plus FSI events (right) generated by the event generator discussed above.

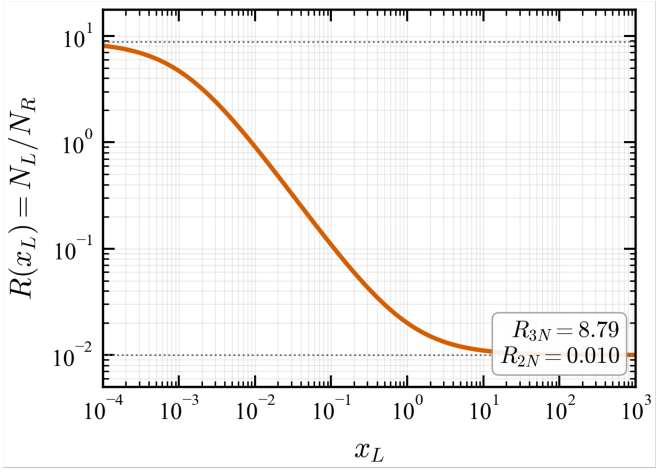


FIG. 6. Ratio of the number of events in the left (N_L) and right (N_R) triangular regions as a function of $x_L \equiv \frac{N_{2N}}{N_{3N}}$; the background-to-signal ratio, defined as the number of 2N-SRC+FSI background events (N_{2N}) relative to the number of 3N-SRC signal events (N_{3N}) in the left region of the phase space (L) shown in Fig. 5.

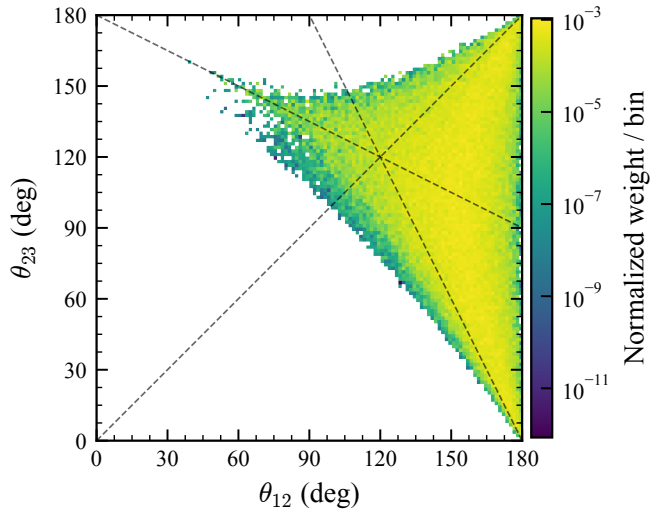


FIG. 7. Opening angle distribution MF plus FSI events generated by the event generator discussed above.

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